THE QUESTION

Which gene programs are shared across all of animal life — and are they the same programs everywhere?

Every animal builds many tissues from **one genome**. The instructions live in **gene co-expression programs** — sets of genes switched on together to make a liver, a neuron, an immune cell.

We asked: across **26 vertebrate species — human to fish, 429 million years** — which of these programs are conserved, and does conservation simply fade with evolutionary distance?

THE APPROACH

Learn the programs once in human, then test them in every other species

1

Find the networks

Human tissue atlas

30 tissues · 3,300 samples

We identify 27 gene co-expression programs — each maps cleanly to a tissue (liver, brain, muscle, immune…).

2

Test conservation

26 vertebrate species

human → fish, 429 My

For each program: does the same gene set still move together in chimp, mouse, chicken, frog, fish?

3

Read the regulators

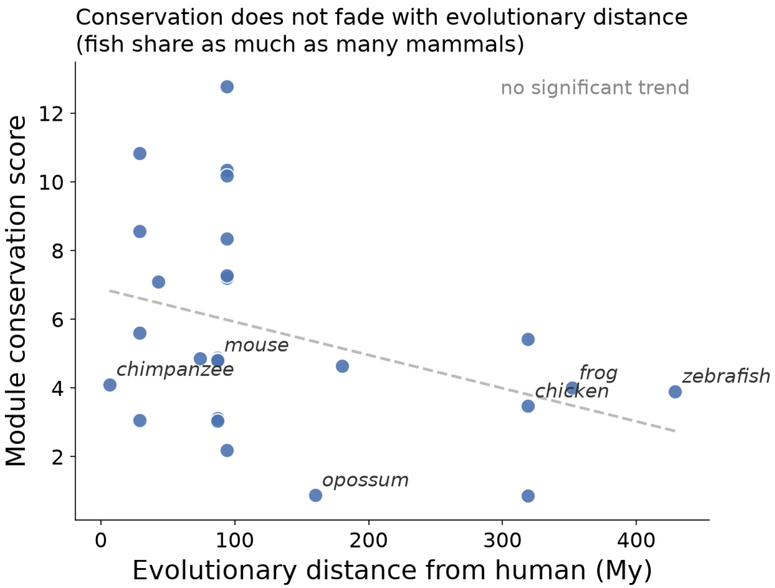
Master-regulator networks per program

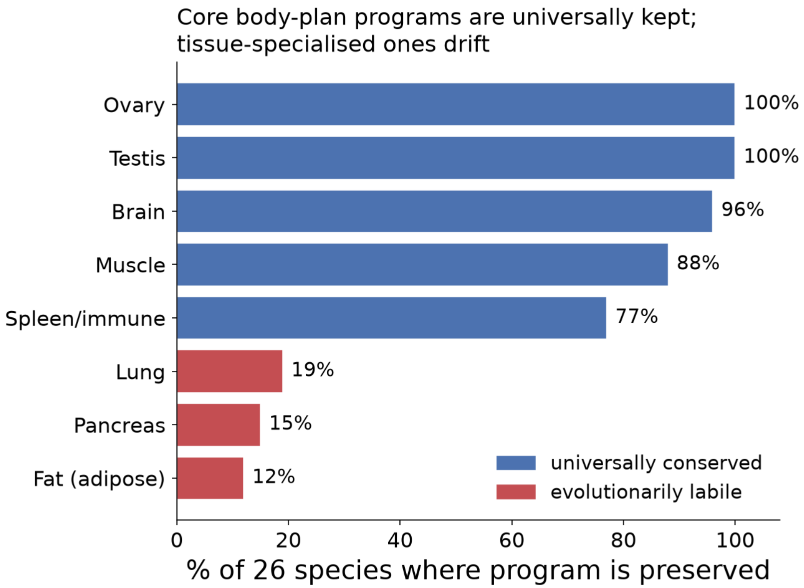
We map the transcription factors that switch each program on — and test whether that logic is conserved too.

Why learn once, then test: programs are discovered in human and checked — not re-derived — in every other species, so a match is genuine conservation, not circular fitting.

KEY FINDING 1 · CONSERVATION IS PROGRAM-DRIVEN

What's conserved depends on the job — not on how distant the species



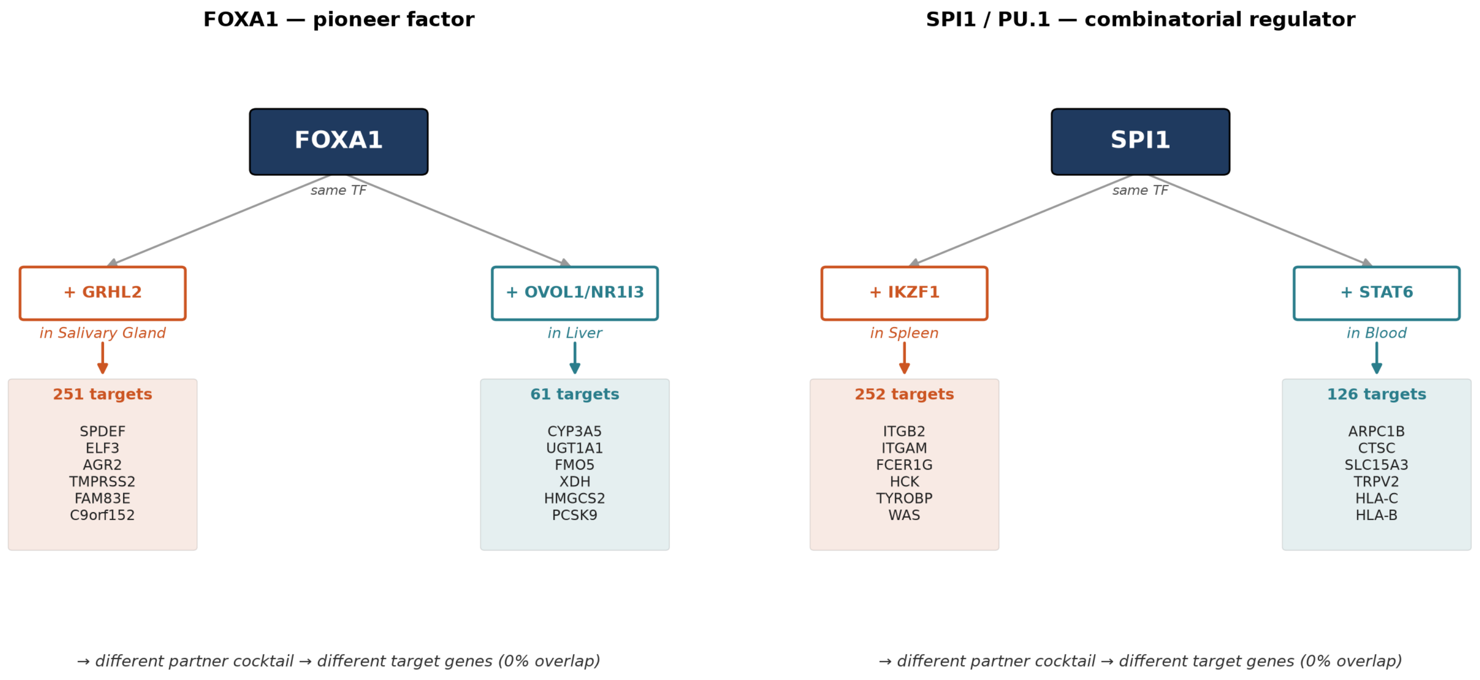


- **Distance doesn't predict conservation**
  A fish can share as much with human as a mouse does.
- **Core programs are universal**
  Brain, testis, ovary, muscle, immune — kept in 77–100% of species.
- **Specialised programs drift**
  Pancreas, lung, fat — re-wired lineage-by-lineage (<20%).
- **Takeaway**
  Evolution locks the ancient core, keeps tinkering with metabolic tissues.

KEY FINDING 2 · THE CONTROL LOGIC IS THE CORE

The same master regulator does different jobs by switching partners

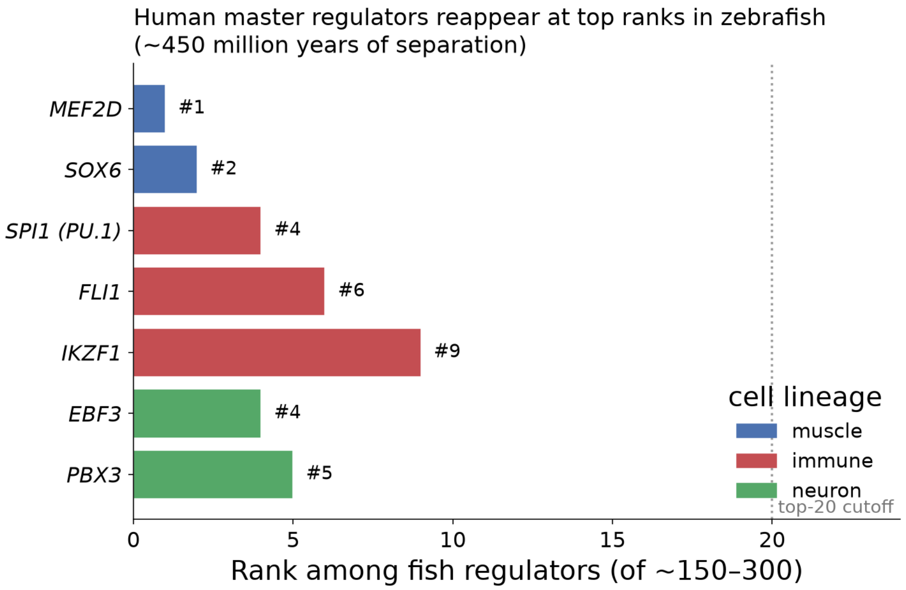
One TF, context-dependent regulons: the partner cocktail selects the targets

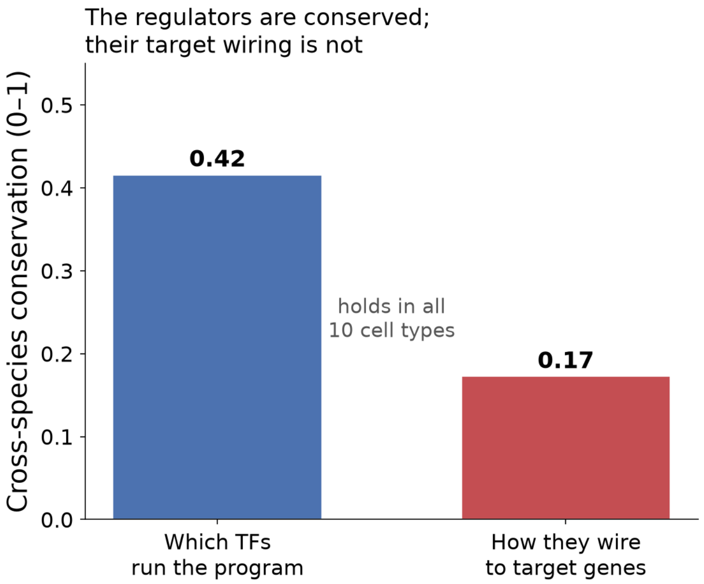


- **One TF, many roles**
  FOXA1 and PU.1 each control different genes in different tissues — partners pick the targets (0% overlap).
- **Regulators are the stable part**
  Which TFs run a program is better conserved than which genes they wire to.
- **Recovered from data alone**
  Textbook combos reappear: liver HNF1A+HNF4A, immune IKZF1+PU.1, the HOX set.
- **Takeaway**
  The conserved architecture carries a conserved control logic.

KEY FINDING 3 · THE CORE REACHES FISH

The regulatory core is ancient — conserved across ~450 million years





- **Master regulators recur in fish**
  Muscle MEF2D #1, immune PU.1 #4, neuron EBF3 #4 in zebrafish ($p<0.02$).
- **Confirmed cell-by-cell**
  Regulator identity conserved (0.42) vs target wiring (0.17), in all 10 cell types.
- **Spans the whole tree**
  Shared to fish means shared across every clade in between.
- **Takeaway**
  One ancient regulatory toolkit underlies all vertebrate tissues, human to fish.